

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

FRIDAY, 20 MAY 2022

TIME : 2.30 PM – 4.30 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are allowed to use a non-scientific calculator.

Answer questions only in the answer booklet provided.

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**SECTION A:** Answer **ALL THREE (3)** questions.

- Q1. (a) Based on the definition in the program below, what is the output of the following cout statements in the table? If the cout statement causes error, then just put "Error" in the output column. Redraw the table to show your answer.

```
#include <iostream>
using namespace std;

struct student {
    char name[30];
    char id[15];
    char course[30];
    int age;
    char gender;

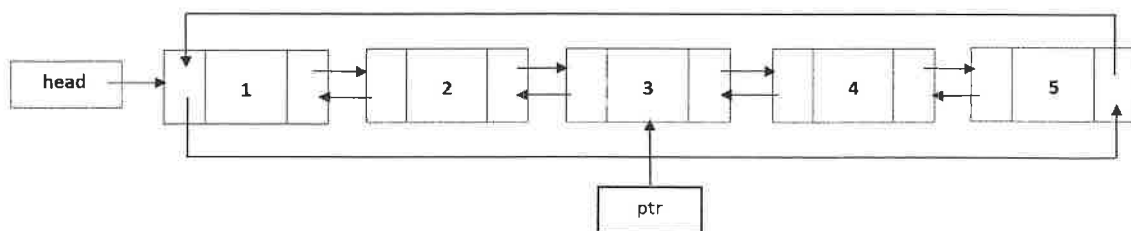
    //name, id, course, age, gender
    student(const char *, const char *, const char *, int, char);
};

int main () {
    student stu("Mary Wong", "22ACB123456", "Computer Science", 20, 'F');
    student *stuptr = &stu;
    return 0;
}
```

Statements	Output
cout << stu.id[6];	
cout << stuptr->course[7];	
cout << stuptr->gender[0];	
cout << stuptr->name;	
cout << stu->age;	
cout << stu.course;	

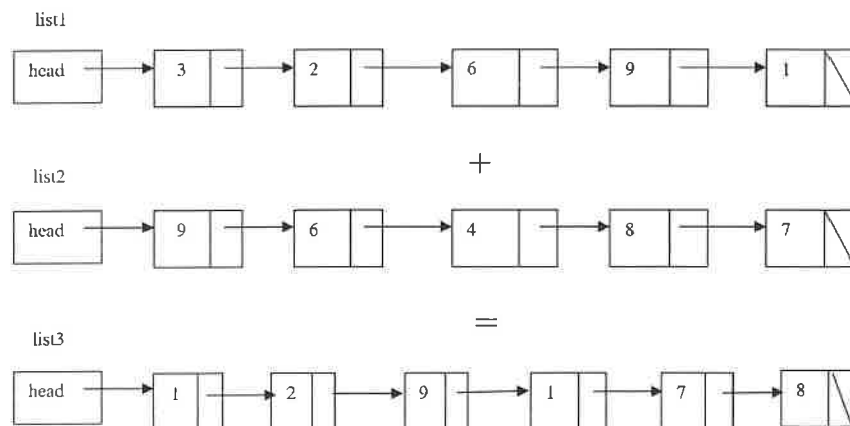
(6 marks)

- (b) A circular doubly linked list with 5 nodes is given in Figure 1 below. The pointer head is pointing to the first node and pointer ptr is pointing to the third node in the list. Answer the following question based on this list.

**Figure 1. Circular Doubly Linked List.**

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q1. (b) (Continued)**

- (i) Write statement(s) to print node with value 3 in the list using the head pointer. (2 marks)
- (ii) Write statement(s) to print node with value 1 in the list using the ptr pointer. (2 marks)
- (iii) Write statement(s) to print the contents of the list in reverse order starting from the last node using the head pointer. You must traverse the list using loop to print the contents. (Note: You can declare other variables if it is needed) (4 marks)
- (iv) Write statement(s) to delete the third node using ptr pointer. The list must still be a circular doubly linked list after the third node is deleted. (3 marks)
- (c) A linked list can be used to represent an integer and we can perform arithmetic operations to this integer stored in the linked list. Figure 2 shows the addition of list1 and list2 where the result is stored in list3. Assume integers in list1 and list2 have the same number of digits. Write a function **bool addition(List list1, List list2, List *list3)** to add the integer stored in list1 and list2 and store the result in list3. The function will return false if list1 or list2 is empty or if list3 is not empty. You need to print the output in the Figure 2 as shown below. Assume list1 is storing 32691 and list2 is storing 96487 in the sample output.

**Figure 2. Contents in list1, list2 and list3.****Sample Output:**

The following is the result of addition operation of two integers in list1 and list2:

$$32691 + 96487 = 129178$$

(8 marks)

[Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

Q2. (a) Answer the following question based on the segment of code below.

```
Stack stack1, stack2;
char s1[10], s2[10];
char arr[10] = { 's', 'y', 'q', 'n', 'a', 'j', 'k', 'b', 'p', 'w' };
int a = 0, b = 0;
char temp1, temp2;

for (int i = 0; i < 10; i++) {
    if (i % 2 == 0)
        stack1.push(arr[i]);
    else
        stack2.push(arr[i]);
}

//*****SECTION A*****

while (!stack1.empty() && !stack2.empty()) {
    if (stack1.pop(temp1) && stack2.pop(temp2)) {
        s1[a++] = temp1;
        s1[a++] = temp2;
    }
    if (stack1.pop(temp1) && stack2.pop(temp2)) {
        s2[b++] = temp1;
        s2[b++] = temp2;
    }
}
s1[a] = '\0';
s2[b] = '\0';
cout << "\ns1: " << s1 << "\n";
cout << "\ns2: " << s2 << "\n";
```

- (i) Draw the content of the stack1 and stack2 after the code is executed up to the comment with “SECTION A”. You need to draw the stack in the linked list format as shown in Figure 2 in Question 1 (c). (6 marks)
- (ii) What is the output of this program? (4 marks)

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVINGQ2. (Continued)

- (b) Answer the following question based on the segment of code below.

<pre>void process(int, Queue) ; void main() { Queue q1; //store int Queue q2; //store int int a; process(33, &q1); cout << "\nQueue 1 Contain: "; while (!q1.empty()) { q1.dequeue(a); cout << a << " "; } process(63, &q2); }</pre> <i>// continue second column</i>	<pre><i>//continue from first column</i> cout << "\nQueue 2 Contain: "; while (!q2.empty()) { q2.dequeue(a); cout << a << " "; } } void process(int x, Queue *q) { if (x == 0) return; else process(x / 2, q); q->enqueue(x % 2); }</pre>
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- (i) Draw the content of Queue 1 **AFTER** the function call **process(33, &q1)** and content of Queue 2 **AFTER** the function call **process(63, &q2)**. You need to draw the queue in the linked list format shown in Figure 2 in Question 1(c) and you need to show which node the head and tail pointers are pointing to. (6 marks)
- (ii) What is the output of this program? (4 marks)
- (c) Write a function **bool reverseQueue(Queue *q)** to reverse the content in a queue using a stack. Refer to the usage of the function in the main() below. This function will return false if the queue is empty and true otherwise. (5 marks)

<pre>int main(){ Queue q; for (int i = 1; i <= 10; i++) q.enqueue(i); cout << "\nQueue Content Before Reverse:\n"; if (!q.print()) cout << "\nCannot print queue.\n"; if (!reverseQueue(&q)) cout << "\nCannot reverse empty queue.\n\n"; else { cout << "\nQueue Content After Reverse:\n"; if (!q.print()) cout << "\nCannot print queue.\n"; } return 0; }</pre>	<u>Sample Output from main():</u> Queue Content Before Reverse: 1 2 3 4 5 6 7 8 9 10 Queue Content After Reverse: 10 9 8 7 6 5 4 3 2 1
--	---

[Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING

- Q3. (a) Consider the unsorted array as below, apply **Merge Sort** technique to sort the following numbers in ascending order. You need to show all the steps of sorting process using diagram. (8 marks)

14	33	27	10	35	19	42	44
----	----	----	----	----	----	----	----

- (b) The following table shows the steps for sorting the sequence of {3, 7, 4, 9, 5, 2, 6, 1} using insertion sort. In each step, the key under consideration is highlighted. The key that was moved in the previous step is marked with an asterisk. Redraw the table and fill in the value for the boxes with X only. (10 marks)

X	7	4	9	5	2	6	1
3*	X	4	9	5	2	6	1
3	7*	X	9	5	2	6	1
3	4*	7	X	5	2	6	1
3	4	7	9*	X	2	6	1
3	4	5*	7	9	X	6	1
2*	3	4	5	7	9	X	1
2	3	4	5	6*	7	9	X
X	X	X	X	X	X	X	X

- (c) Program below shows a code that the outer loop will run for n number of iterations. In each iteration of the outer loop, the inner loop will run for n iterations of their own. State and justify the time complexity of the code. (3 marks)

```
int fun(int n) {
    int i = 0, j = 0, k = 0, m = 0;
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            for (k = 0; k < n; k++)
                m += 1;
    return m;
}
```

- (d) How many recursive calls are made when computing the sum of the list [2,4,6,8,10]? Justify your answer. (4 marks)
- [Total : 25 marks]

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**SECTION B:** Answer **ONLY ONE (1)** question.

- Q4. (a) Draw the Binary Search Tree (BST) after the following integer values are inserted to an empty BST. (6 marks)

9, 7, 14, 3, 18, 4, 13, 19, 16

- (b) What is the purpose of the function below? Redraw the Binary Search Tree 1 below if the function is called with node pointing to the root in Binary Search Tree 1 shown in Figure 3.

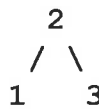


Figure 3. Binary Search Tree 1.

```
void BST<T>::TreeOperation(BTNode<T> *node) {
```

```
    BTNode<T> *oldLeft;
```

```
    if (node==NULL)
        return;
```

```
    TreeOperation(node->left);
    TreeOperation (node->right);
```

```
    oldLeft = node->left;
    node->left = new BTNode<T>(node->item);
    node->left->left = oldLeft;
```

```
}
```

(6 marks)

- (c) Answer question c (i) and c (ii) using the Binary Tree given in Figure 4.

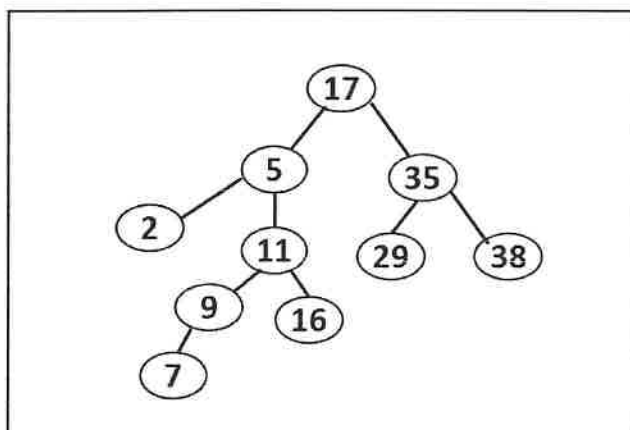


Figure 4. Binary tree.

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q4. (c) (Continued)**

- (i) First, determine whether the tree drawn in Figure 4 is balance or not if the node 16 is deleted. After that, specify which node is the deepest unbalance node and the case of node deletion (e.g. RR, RL ...). (3 marks)
- (ii) Balance the tree after the node 16 is deleted in question 4(c)(i) and then redraw the resultant tree. (3 marks)
- (d) The Tower of Hanoi is a mathematical puzzle which solution illustrates recursion. Given the function definition as follows:

$$\text{hanoi}(n) = \begin{cases} 1 & \text{if } n = 1 \\ 2 \cdot \text{hanoi}(n - 1) + 1 & \text{if } n > 1 \end{cases}$$

Compute the recurrence relation for $n = 4$. You need to show the step by step computation to define your final answer. (7 marks)

[Total : 25 marks]

- Q5. (a) Figure 5 shows an Undirected Graph A.

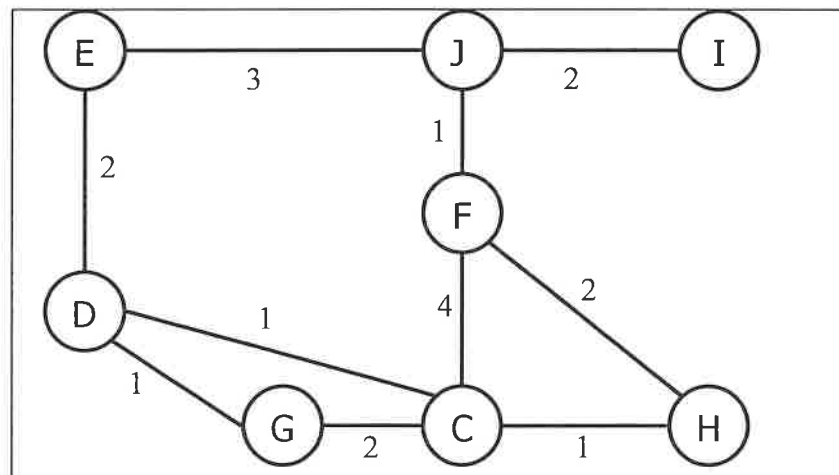
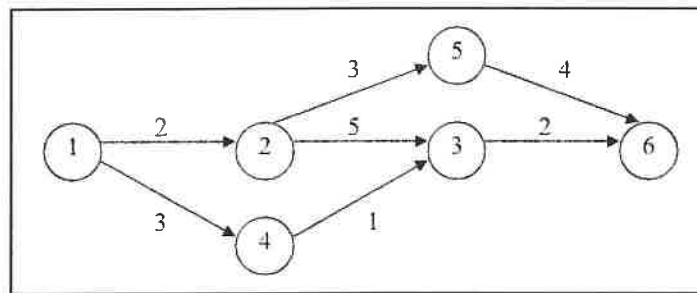


Figure 5. Undirected Graph A.

- (i) Construct an adjacency matrix representation of the Undirected Graph A in Figure 5. (4 marks)
- (ii) Starting at vertex C, traverse the graph by using **Depth-First Search (DFS)**. The vertices are to be traversed alphabetically. Give the order in which the vertices are traversed. (4 marks)

UCCD1024 DATA STRUCTURE AND ALGORITHMIC PROBLEM SOLVING**Q5. (a) (Continued)**

- (iii) Starting at vertex C, traverse the graph by using **Breadth-First Search (BFS)**. The vertices are to be traversed alphabetically. Give the order in which the vertices are traversed. (4 marks)
- (b) Write a Binary Search function using **int BinarySearchRec (int key, int list [], int start, int end)** to perform Binary Search using **recursion**. The variable **key** is the value to search and **list** is the array to be searched. The **start** and **end** indicate the area to search in the array. The function will return -1 if the key is not found. (7 marks)
- (c) Consider the following directed graph shown in Figure 6.

**Figure 6. Directed Graph.**

Referring to the directed graph shown in Figure 6, suppose that we are using the Dijkstra's algorithm to determine the shortest paths from vertex 1 to every other vertex. Provide the list of travelled distances that the algorithm will be processed from the vertex 1 to each of the other 5 vertices. Table 1 provides the hints for the list of travelled distances. (6 marks)

Table 1: Predecessor vertex and Successor vertex of each vertex travelled.

Predecessor vertex	Successor vertex
N/A	1
1	2
4	3
1	4
2	5
3	6

[Total : 25 marks]

